MSG No. 73.553 I_cbca [Type IV, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `2'22'`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	`[1,3,18,28]`
2	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[2,12,17,19]`
3	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[4,10,25,27]`
4	$\left[\frac{3}{4}, 0, 0\right]$	`[5,7,22,32]`
5	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[6,16,21,23]`
6	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[8,14,29,31]`
7	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,11,20,26]`
8	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,15,24,30]`

Table 2: Wyckoff site: `8b`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[1,2,19,28]`
2	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[3,12,17,18]`
3	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[4,11,25,26]`
4	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[5,6,23,32]`
5	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[7,16,21,22]`
6	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[8,15,29,30]`
7	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[9,10,20,27]`
8	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[13,14,24,31]`

Table 3: Wyckoff site: `8c`, site symmetry: `2'/m'..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,18,22]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[2,6,17,21]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[3,7,28,32]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[4,8,27,31]`
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,13,26,30]`
6	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10,14,25,29]`
7	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[11,15,20,24]`
8	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[12,16,19,23]`

Table 4: Wyckoff site: `8d`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,21,22]`
2	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3,12,23,32]`
3	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[4,11,24,31]`
4	$\left[0, 0, \frac{3}{4}\right]$	`[5,6,17,18]`
5	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[7,16,19,28]`

continued ...

Table 4

No.	position	mapping
6	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[8,15,20,27]
7	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[9,10,29,30]
8	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[13,14,25,26]

Table 5: Wyckoff site: 8e, site symmetry: .2/m'.

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[1,3,29,31]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[2,12,24,30]
3	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[4,10,22,32]
4	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[5,7,25,27]
5	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[6,16,20,26]
6	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[8,14,18,28]
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[9,11,21,23]
8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[13,15,17,19]

Table 6: Wyckoff site: 8f, site symmetry: .2'/m'.

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,13,19,31]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2,14,24,28]
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,15,17,29]
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[4,16,22,26]
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[5,9,23,27]
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[6,10,20,32]
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[7,11,21,25]
8	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[8,12,18,30]

Table 7: Wyckoff site: 8g, site symmetry: m'm'2

No.	position	mapping
1	$\left[0, \frac{1}{4}, z\right]$	[1,4,22,31]
2	$\left[0, \frac{3}{4}, \frac{1}{2} - z\right]$	[2,11,21,24]
3	$\left[\frac{1}{2}, \frac{1}{4}, -z\right]$	[3,10,29,32]
4	$\left[0, \frac{3}{4}, -z\right]$	[5,8,18,27]
5	$\left[0, \frac{1}{4}, z + \frac{1}{2}\right]$	[6,15,17,20]
6	$\left[\frac{1}{2}, \frac{3}{4}, z\right]$	[7,14,25,28]
7	$\left[\frac{1}{2}, \frac{3}{4}, z + \frac{1}{2}\right]$	[9,12,23,30]
8	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2} - z\right]$	[13,16,19,26]

Table 8: Wyckoff site: `16h`, site symmetry: `2'..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,18]`
2	$[x,\ 0,\ \frac{1}{2}]$	`[2,17]`
3	$[\frac{1}{2}-x,\ 0,\ 0]$	`[3,28]`
4	$[-x,\ \frac{1}{2},\ 0]$	`[4,27]`
5	$[-x,\ 0,\ 0]$	`[5,22]`
6	$[-x,\ 0,\ \frac{1}{2}]$	`[6,21]`
7	$[x+\frac{1}{2},\ 0,\ 0]$	`[7,32]`
8	$[x,\ \frac{1}{2},\ 0]$	`[8,31]`
9	$[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}]$	`[9,26]`
10	$[x+\frac{1}{2},\ \frac{1}{2},\ 0]$	`[10,25]`
11	$[-x,\ \frac{1}{2},\ \frac{1}{2}]$	`[11,20]`
12	$[\frac{1}{2}-x,\ 0,\ \frac{1}{2}]$	`[12,19]`
13	$[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}]$	`[13,30]`
14	$[\frac{1}{2}-x,\ \frac{1}{2},\ 0]$	`[14,29]`
15	$[x,\ \frac{1}{2},\ \frac{1}{2}]$	`[15,24]`
16	$[x+\frac{1}{2},\ 0,\ \frac{1}{2}]$	`[16,23]`

Table 9: Wyckoff site: `16i`, site symmetry: `2..`

No.	position	mapping
1	$[x,\ 0,\ \frac{1}{4}]$	`[1,2]`
2	$[\frac{1}{2}-x,\ 0,\ \frac{3}{4}]$	`[3,12]`
3	$[-x,\ \frac{1}{2},\ \frac{1}{4}]$	`[4,11]`
4	$[-x,\ 0,\ \frac{3}{4}]$	`[5,6]`
5	$[x+\frac{1}{2},\ 0,\ \frac{1}{4}]$	`[7,16]`
6	$[x,\ \frac{1}{2},\ \frac{3}{4}]$	`[8,15]`
7	$[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}]$	`[9,10]`
8	$[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}]$	`[13,14]`
9	$[x,\ 0,\ \frac{3}{4}]$	`[17,18]`
10	$[\frac{1}{2}-x,\ 0,\ \frac{1}{4}]$	`[19,28]`
11	$[-x,\ \frac{1}{2},\ \frac{3}{4}]$	`[20,27]`
12	$[-x,\ 0,\ \frac{1}{4}]$	`[21,22]`
13	$[x+\frac{1}{2},\ 0,\ \frac{3}{4}]$	`[23,32]`
14	$[x,\ \frac{1}{2},\ \frac{1}{4}]$	`[24,31]`
15	$[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}]$	`[25,26]`
16	$[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}]$	`[29,30]`

Table 10: Wyckoff site: `16j`, site symmetry: `.2.`

No.	position	mapping
1	$[\frac{1}{4},\ y,\ 0]$	`[1,3]`
2	$[\frac{1}{4},\ -y,\ \frac{1}{2}]$	`[2,12]`
3	$[\frac{3}{4},\ \frac{1}{2}-y,\ 0]$	`[4,10]`
4	$[\frac{3}{4},\ -y,\ 0]$	`[5,7]`

continued ...

Table 10

No.	position	mapping
5	$\left[\frac{3}{4}, y, \frac{1}{2}\right]$	[6,16]
6	$\left[\frac{1}{4}, y+\frac{1}{2}, 0\right]$	[8,14]
7	$\left[\frac{3}{4}, y+\frac{1}{2}, \frac{1}{2}\right]$	[9,11]
8	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{2}\right]$	[13,15]
9	$\left[\frac{1}{4}, y, \frac{1}{2}\right]$	[17,19]
10	$\left[\frac{1}{4}, -y, 0\right]$	[18,28]
11	$\left[\frac{3}{4}, \frac{1}{2}-y, \frac{1}{2}\right]$	[20,26]
12	$\left[\frac{3}{4}, -y, \frac{1}{2}\right]$	[21,23]
13	$\left[\frac{3}{4}, y, 0\right]$	[22,32]
14	$\left[\frac{1}{4}, y+\frac{1}{2}, \frac{1}{2}\right]$	[24,30]
15	$\left[\frac{3}{4}, y+\frac{1}{2}, 0\right]$	[25,27]
16	$\left[\frac{1}{4}, \frac{1}{2}-y, 0\right]$	[29,31]

Table 11: Wyckoff site: 16k, site symmetry: .2'.

No.	position	mapping
1	$\left[\frac{1}{4}, y, \frac{1}{4}\right]$	[1,19]
2	$\left[\frac{1}{4}, -y, \frac{1}{4}\right]$	[2,28]
3	$\left[\frac{1}{4}, y, \frac{3}{4}\right]$	[3,17]
4	$\left[\frac{3}{4}, \frac{1}{2}-y, \frac{1}{4}\right]$	[4,26]
5	$\left[\frac{3}{4}, -y, \frac{3}{4}\right]$	[5,23]
6	$\left[\frac{3}{4}, y, \frac{3}{4}\right]$	[6,32]
7	$\left[\frac{3}{4}, -y, \frac{1}{4}\right]$	[7,21]
8	$\left[\frac{1}{4}, y+\frac{1}{2}, \frac{3}{4}\right]$	[8,30]
9	$\left[\frac{3}{4}, y+\frac{1}{2}, \frac{3}{4}\right]$	[9,27]
10	$\left[\frac{3}{4}, \frac{1}{2}-y, \frac{3}{4}\right]$	[10,20]
11	$\left[\frac{3}{4}, y+\frac{1}{2}, \frac{1}{4}\right]$	[11,25]
12	$\left[\frac{1}{4}, -y, \frac{3}{4}\right]$	[12,18]
13	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{4}\right]$	[13,31]
14	$\left[\frac{1}{4}, y+\frac{1}{2}, \frac{1}{4}\right]$	[14,24]
15	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{3}{4}\right]$	[15,29]
16	$\left[\frac{3}{4}, y, \frac{1}{4}\right]$	[16,22]

Table 12: Wyckoff site: 16l, site symmetry: ..2'

No.	position	mapping
1	$\left[\frac{1}{4}, 0, z\right]$	[1,28]
2	$\left[\frac{1}{4}, 0, \frac{1}{2}-z\right]$	[2,19]
3	$\left[\frac{1}{4}, 0, -z\right]$	[3,18]
4	$\left[\frac{3}{4}, \frac{1}{2}, z\right]$	[4,25]
5	$\left[\frac{3}{4}, 0, -z\right]$	[5,32]
6	$\left[\frac{3}{4}, 0, z+\frac{1}{2}\right]$	[6,23]
7	$\left[\frac{3}{4}, 0, z\right]$	[7,22]
8	$\left[\frac{1}{4}, \frac{1}{2}, -z\right]$	[8,29]

continued ...

Table 12

No.	position	mapping
9	$\left[\frac{3}{4}, \frac{1}{2}, z+\frac{1}{2}\right]$	[9,20]
10	$\left[\frac{3}{4}, \frac{1}{2}, -z\right]$	[10,27]
11	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}-z\right]$	[11,26]
12	$\left[\frac{1}{4}, 0, z+\frac{1}{2}\right]$	[12,17]
13	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}-z\right]$	[13,24]
14	$\left[\frac{1}{4}, \frac{1}{2}, z\right]$	[14,31]
15	$\left[\frac{1}{4}, \frac{1}{2}, z+\frac{1}{2}\right]$	[15,30]
16	$\left[\frac{3}{4}, 0, \frac{1}{2}-z\right]$	[16,21]

Table 13: Wyckoff site: 16m, site symmetry: m'..

No.	position	mapping
1	$[0, y, z]$	[1,22]
2	$\left[0, -y, \frac{1}{2}-z\right]$	[2,21]
3	$\left[\frac{1}{2}, y, -z\right]$	[3,32]
4	$\left[0, \frac{1}{2}-y, z\right]$	[4,31]
5	$[0, -y, -z]$	[5,18]
6	$\left[0, y, z+\frac{1}{2}\right]$	[6,17]
7	$\left[\frac{1}{2}, -y, z\right]$	[7,28]
8	$\left[0, y+\frac{1}{2}, -z\right]$	[8,27]
9	$\left[\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[9,30]
10	$\left[\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[10,29]
11	$\left[0, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[11,24]
12	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	[12,23]
13	$\left[\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[13,26]
14	$\left[\frac{1}{2}, y+\frac{1}{2}, z\right]$	[14,25]
15	$\left[0, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[15,20]
16	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	[16,19]

Table 14: Wyckoff site: 16n, site symmetry: .m'.

No.	position	mapping
1	$\left[x, \frac{1}{4}, z\right]$	[1,31]
2	$\left[x, \frac{3}{4}, \frac{1}{2}-z\right]$	[2,24]
3	$\left[\frac{1}{2}-x, \frac{1}{4}, -z\right]$	[3,29]
4	$\left[-x, \frac{1}{4}, z\right]$	[4,22]
5	$\left[-x, \frac{3}{4}, -z\right]$	[5,27]
6	$\left[-x, \frac{1}{4}, z+\frac{1}{2}\right]$	[6,20]
7	$\left[x+\frac{1}{2}, \frac{3}{4}, z\right]$	[7,25]
8	$\left[x, \frac{3}{4}, -z\right]$	[8,18]
9	$\left[x+\frac{1}{2}, \frac{3}{4}, z+\frac{1}{2}\right]$	[9,23]
10	$\left[x+\frac{1}{2}, \frac{1}{4}, -z\right]$	[10,32]
11	$\left[-x, \frac{3}{4}, \frac{1}{2}-z\right]$	[11,21]
12	$\left[\frac{1}{2}-x, \frac{3}{4}, z+\frac{1}{2}\right]$	[12,30]

continued ...

Table 14

No.	position	mapping
13	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[13,19]
14	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ z\right]$	[14,28]
15	$\left[x,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[15,17]
16	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[16,26]

Table 15: Wyckoff site: 32o, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[2]
3	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	[3]
4	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[6]
7	$\left[x+\frac{1}{2},\ -y,\ z\right]$	[7]
8	$\left[x,\ y+\frac{1}{2},\ -z\right]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[10]
11	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[14]
15	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[16]
17	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[17]
18	$[x,\ -y,\ -z]$	[18]
19	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[19]
20	$\left[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[20]
21	$\left[-x,\ -y,\ \frac{1}{2}-z\right]$	[21]
22	$[-x,\ y,\ z]$	[22]
23	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[23]
24	$\left[x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[24]
25	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[25]
26	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[26]
27	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	[27]
28	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	[28]
29	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$	[29]
30	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[30]
31	$\left[x,\ \frac{1}{2}-y,\ z\right]$	[31]
32	$\left[x+\frac{1}{2},\ y,\ -z\right]$	[32]